

React Hydration Error Fix - Signup Page

Problem

The signup page was experiencing a React Hydration Error with the message:

“Hydration failed because the initial UI does not match what was rendered on the server.”

The error was specifically related to Clerk’s `<SignIn>` and `<Portal>` components, indicating a server-side and client-side rendering mismatch.

Root Causes

1. useSearchParams() without Suspense Boundary

- The `useSearchParams()` hook from Next.js requires a Suspense boundary to prevent hydration mismatches
- During server-side rendering, search params may not be available, causing differences between server and client renders
- **Solution:** Wrapped the component in a `<Suspense>` boundary with a loading fallback

2. Theme Detection Mismatch

- The `useTheme()` hook from `next-themes` reads from `localStorage`, which is only available on the client
- Server renders with undefined/default theme, while client renders with the actual theme from `localStorage`
- This causes the initial HTML from the server to differ from what React expects on the client
- **Solution:**
 - Added `isMounted` state to defer theme-dependent rendering until after hydration
 - Shows a loading spinner during the initial mount to prevent mismatch
 - Properly handles `system` theme by resolving it to the actual theme value

Changes Made

File: `/app/(auth)/signup/[...signup]/page.tsx`

1. Added Suspense Import

```
tsx
import { useState, useEffect, useCallback, Suspense } from "react";
```

2. Created SignUpContent Component

- Renamed the main component to `SignUpContent`
- This inner component contains all the logic and hooks

3. Added Mounted State

```
``tsx
const [isMounted, setIsMounted] = useState(false);
```

```
useEffect(() => {
  setIsMounted(true);
}, []);
```

```

### 1. Added Loading State for Hydration

```
tsx
 if (!isMounted) {
 return (
 <div className="flex flex-col items-center w-full max-w-md">
 <div className="w-full p-6 border border-gray-200 rounded-lg shadow-md bg-white
text-center">
 <Loader2 className="animate-spin h-8 w-8 mx-auto text-purple-600" />
 </div>
 </div>
);
 }
}
```

### 2. Fixed Theme Usage

```
```tsx
const { theme, systemTheme } = useTheme();
const currentTheme = theme === "system" ? systemTheme : theme;

// Used currentTheme instead of theme in SignUp component
```
```

### 1. Wrapped in Suspense Boundary

```
tsx
export default function SignUpPage() {
 return (
 <Suspense fallback={
 <div className="flex flex-col items-center w-full max-w-md">
 <div className="w-full p-6 border border-gray-200 rounded-lg shadow-md bg-white
text-center">
 <Loader2 className="animate-spin h-8 w-8 mx-auto text-purple-600" />
 </div>
 </div>
 }>
 <SignUpContent />
 </Suspense>
);
}
```

## Benefits

1. **Eliminates Hydration Errors:** Server and client renders now match perfectly
2. **Maintains Functionality:** All existing features (email pre-fill, profile claiming, etc.) work as expected
3. **Better User Experience:** Shows appropriate loading states during initialization

4. **Next.js 13+ Best Practices:** Follows official Next.js documentation for handling `useSearchParams()`
5. **Theme Consistency:** Properly handles theme switching without causing hydration issues

## Testing Recommendations

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1. **Test with different URL parameters:**

- `/signup` (no params)
- `/signup?email=test@example.com` (with email)
- `/signup?email=test@example.com&token=abc123&payment=success` (full params)

2. **Test theme switching:**

- Light theme
- Dark theme
- System theme

3. **Check browser console:**

- Should see no hydration warnings
- Should see no React errors

4. **Verify functionality:**

- Email pre-fill works
- Profile claiming works after signup
- Redirects work properly

## References

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- [Next.js useSearchParams Documentation](https://nextjs.org/docs/app/api-reference/functions/use-search-params) (<https://nextjs.org/docs/app/api-reference/functions/use-search-params>)
- [React Hydration Documentation](https://react.dev/reference/react-dom/client/hydrateRoot) (<https://react.dev/reference/react-dom/client/hydrateRoot>)
- [next-themes GitHub](https://github.com/pacocoursey/next-themes) (<https://github.com/pacocoursey/next-themes>)